

Susanta Khamrui

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IIT Kharagpur, West Bengal, India

Professional Profile

PhD Scholar in Computer Science and Engineering at IIT Kharagpur, specializing in Neuro-symbolic AI, Reasoning, and Verification in Large Language Models (LLMs). Building upon a rigorous dual-foundation in Digital Signal Processing and Deep Learning (M.Tech, IIITDM Jabalpur), my current research explores the intersection of neural representation learning and symbolic logic to enable transparent, factual, and reliable AI systems. Proficient in Python, PyTorch, and optimization mathematics, I am highly motivated to design intelligent frameworks that bridge the gap between raw perceptual data and complex, verifiable reasoning tasks.

Technical Interests

- **Neuro-symbolic AI & Reasoning:** Unifying statistical learning with symbolic verification to enable knowledge-guided inference. Exploring external neuro-symbolic verifiers and structured knowledge to actively steer model generation and scale test-time compute.
- **Dynamic Verification in LLMs:** Advancing beyond standard post-hoc verification by designing dynamic verifiers that enable branching and backtracking correction during inference. Extending verification capabilities to ambiguous natural language for high-stakes domains (e.g., healthcare, law).
- **Representation Learning:** Extracting robust, isolated latent embeddings from complex, noisy multi-modal data to support high-level abstract reasoning and cross-modal alignment.
- **Vision & Language Modeling:** Bridging visual perception with natural language understanding to build robust systems capable of spatial reasoning and complex visual question answering.

Education

- **Indian Institute of Technology (IIT) Kharagpur** **July 2026 – Present**
Ph.D. in Computer Science and Engineering *Supervisor: Dr. Somak Aditya*
Research Area: Neuro-symbolic AI, Reasoning, and Verification in LLMs
- **PDPM IIITDM Jabalpur** **August 2024 – June 2026**
M.Tech in Computer Science and Engineering (Data Science Specialization) *CGPA: 8.3 / 10*
Thesis: Representation Learning via Hybrid Fusion Architectures for Visual Segmentation
- **Dr. B. C. Roy Engineering College, Durgapur (MAKAUT)** **August 2019 – June 2023**
B.Tech in Electronics and Communication Engineering *CGPA: 8.5 / 10*
Focus: Digital Signal Processing, Probability & Statistics, and Engineering Mathematics

Research & Project Experience

- **Neuro-symbolic Reasoning & LLM Verification (PhD Research)** *July 2026 – Present*
Supervisor: Dr. Somak Aditya, IIT Kharagpur
 - **The Focus:** Investigating methodologies to combine neural networks with logical systems to perform complex, multi-hop reasoning. The goal is to address the black-box nature of Large Language Models (LLMs) by injecting structured knowledge and verifying reasoning traces.
 - **Current Work:** Developing neuro-symbolic pipelines to evaluate and enhance the reasoning capabilities of Transformer-based models. Focusing on conditional and mathematical reasoning, evaluating competency through ontology-guided perturbations, and building safety verification frameworks for next-generation AI agents.
- **Advanced Representation Learning (M.Tech Thesis)** *February 2025 – June 2026*
Supervisor: Dr. Ayan Seal, PDPM IIITDM Jabalpur
 - **The Problem:** Medical vision models frequently struggle to generalize and extract accurate representations when exposed to severe background noise, class imbalance, and data scarcity.

- **My Solution:** Engineered "TAP-FuseNet", a novel hybrid architecture. Utilized mathematical optimization and Transformer attention mechanisms to force the network to focus on relevant structural details over background noise.
- **What I Learned:** Developed deep, hands-on expertise in PyTorch and extracting isolated features from chaotic visual data—a skill set directly transferable to learning causally grounded concepts from unstructured data for neuro-symbolic reasoning.

Publications

- S. Khamrui and A. Seal, *ADPNet: An Attention-Driven Polyp Segmentation Network with Statistical Feature Extraction*.
7th International Conference on Frontiers in Computing and Systems (COMSYS), 2026. (Accepted)
- S. Khamrui, A. Seal, A. K. Gupta, M. Penhaker, and O. Krejcar, *An Instrument for Small Polyp Localization in Colonoscopy Images using a Task-Aware Progressive Fusion Network*.
(Under Review)
- S. Khamrui, A. Seal, M. Penhaker, and O. Krejcar, *BAHNet: A Boundary-Aware Hybrid Network for Small Polyp Segmentation in Colonoscopy*, 2026.
(Under Review)

Certifications & Training

- **Security and Threats of Generative AI (GIAN)** *December 2025*
Coursework on Model Robustness, Adversarial Defense, and Statistical Reliability of Deep Learning models.
- **Summer School on Deep Learning** *June 2025*
Intensive training on Transformer architectures, Vision Transformers (ViT), and advanced optimization techniques.

Technical Skills

- **Languages:** Python, C++, Java.
- **Machine Learning & DL:** PyTorch, LLMs, Transformers, Neuro-symbolic pipelines.
- **Tools & Workflows:** Git/GitHub, Linux, LaTeX, HPC cluster computing, VS Code.

Referees

Dr. Somak Aditya (PhD Supervisor)

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Dr. Ayan Seal (M.Tech Supervisor)

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